



Supplementary Materials for

U-Series Dating of Paleolithic Art in 11 Caves in Spain

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Supplementary Materials:

Materials and Methods

The Uranium-series disequilibrium method

The U-series disequilibrium method is based on the radioactive decay of radionuclides within the naturally occurring decay chains. There are three such decay chains, each starts with an actinide nuclide (i.e., ^{238}U , ^{235}U , and ^{232}Th) having a long half live (all have $T_{1/2} > 7 \times 10^8$ years) and ultimately ends with different stable isotopes of lead. For dating speleothems, we make use of an initial elemental fractionation between Th and U in the ^{238}U decay series when carbonate bedrock is dissolved. Differential solubility between uranium and its long lived daughter isotope ^{230}Th means that calcite precipitates (e.g. stalagmites, stalactites and flowstones) contain traces of uranium but, in theory, no ^{230}Th . Over time, there is ingrowth of ^{230}Th from the radioactive decay of ^{238}U until radioactive equilibrium is reached where all isotopes in the series are decaying at the same rate. It is the degree of disequilibrium (measured as $^{230}\text{Th}/^{238}\text{U}$ activity ratio) that can be used together with the activity ratio of the two U isotopes $^{234}\text{U}/^{238}\text{U}$ to calculate the age of the calcite precipitation. Natural processes usually also cause a disequilibrium between ^{238}U and ^{234}U , so the age since formation of a calcite sample is calculated iteratively from measurements of $^{234}\text{U}/^{238}\text{U}$ and $^{230}\text{Th}/^{238}\text{U}$ (36).

An additional problem is the incorporation of detritus in the precipitating calcite. This can be from wind-blown or waterborne sediments. Detrital sediments will bring U and Th and usually will result in the apparent age of a contaminated sample to be an overestimate of the true age. However, the presence of the common thorium isotope, ^{232}Th , indicates the presence of contamination, and there are several methods to correct the U-series date for it. An indication of the degree of detrital contamination is expressed as $^{230}\text{Th}/^{232}\text{Th}$ activity, with high values (>20) indicating little or no effect on the calculated date and low values (<20) indicating the correction on the date will be significant. For very low values of $^{230}\text{Th}/^{232}\text{Th}$ (i.e. <5), the calculated age will be dominated by the assumptions used to correct for the detritus, so we employ two different correction strategies. For samples with minor and moderate levels ($^{230}\text{Th}/^{232}\text{Th} > 5$) of contamination, we correct using an assumed detrital activity ratio of $^{232}\text{Th}/^{238}\text{U} = 1.250 \pm 0.625$, typical of upper crustal silicates (37) and assume ^{230}Th and U isotopes are in equilibrium (i.e. $^{230}\text{Th}/^{238}\text{U} = 1.0$; $^{234}\text{U}/^{238}\text{U} = 1.0$). Note the conservative error on this assumption. For samples with high levels of detritus, we attempt to measure typical detrital $^{230}\text{Th}/^{232}\text{U}$ on the insoluble residues from the calcite samples and report both a date using a crustal silicate correction and a date using the measured detrital value (see for example samples O-21 and O-48). While the date obtained using measured detritus values agrees within error in both cases with the date using an average crustal silicate, we must be cautious in using dates corrected using the insoluble detritus. Soluble Th in the detritus, or the adsorption of authigenic Th onto the detritus may introduce uncertainty in determining the true detrital $^{230}\text{Th}/^{232}\text{Th}$ (38). This uncertainty is unlikely to be as large as the 50% uncertainty we are using for our assumed detrital value, but it is likely to be greater than the uncertainty we give. To be cautious therefore, we base our interpretation of the dates for samples O-21 and O-48 on dates corrected with our assumed rather than measured detrital value.

Method

In order to be certain that a minimum or maximum age is obtained, it is essential to select calcite deposits that have an unambiguous stratigraphic relationship with the painting or engraving. Each potential location was inspected with a hand lens and locations were sampled only where the painting was clearly covered with calcite, or where the underlying calcite was accessible. The calcite deposits ranged from small discontinuous nodules (e.g. Fig. S6), 'trickles' of flowstone (e.g. Fig. S3), through to the occasional continuous flowstone crust. Powdery and soft deposits were avoided. The majority

of samples were removed by scraping with a scalpel, catching the scrapings in a cleaned plastic tray. The calcite was removed in spits, creating aliquots of sample. This allowed regular inspection of the scraped surface and the aliquots of calcite in order to (a) avoid unintended inclusion of scraped pigment, which would contaminate the sample and (b) make sure the sample removed was still entirely from above the painting, so as not to damage it. Aliquots contaminated with pigment or visible detritus were discarded and the remaining aliquots combined to give sample masses of 10-100mg. Where a sufficient thickness of calcite was present (>2mm), 2 samples were removed, representing the upper and lower portions of the calcite crust. In all cases, the dates of these fall in the correct stratigraphic order, demonstrating the integrity of the calcite (Fig. S1). In some cases where the formation was stalactitic, samples were cut with a diamond cutting wheel, or drilled with a carbide drill bit. A further demonstration of the reliability of the technique comes from the distribution of results, which show that the formation ages for calcite on top of art fall between a few hundred years and 40.8 ka (Fig. 2). Since calcite formation has been ongoing in most caves over a period beyond the limit of the U-Th method (c. 500 ka), this distribution would not be expected if the stratigraphic relationship between the art and the calcite was insecure.

Samples were initially inspected under a low power microscope and, where possible, any obvious particles of detritus were removed. The sample was weighed in a Teflon beaker. A few drops of milliQ 18M Ω water were added, and the sample was dissolved by further stepwise addition of 7N HNO₃. A mixed ²²⁹Th/²³⁶U spike was added and left for a few hours to equilibrate. Where appropriate, any insoluble residue was removed by centrifuge. The sample solution was dried by placing the beaker on a hotplate. When nearly dry the sample was treated with 100 μ l 6N HCl and 55 μ l H₂O₂ and left until dry. Finally, the sample was re-dissolved in 600 μ l 6N HCl ready for the ion exchange columns.

U and Th were separated from the sample matrix using ion exchange chromatography and a two column procedure (9). The first column separates U from Th and the second purifies the two fractions. We use 600 μ l of pre-cleaned Bio Rad AG1x8 resin. The sample is introduced into the first column in 6N HCl. The Th fraction is collected immediately as it passes directly through the column. U is then eluted using 1N HBr followed by 18M Ω water. After drying down the two fractions were redissolved in 7N HNO₃ and separately passed down the column for purification. Th is eluted with 6N HCl and U is eluted with 1N HBr. The elutants were dried then redissolved in 0.6N HCl ready for analysis.

U and Th isotope measurements were undertaken using a ThermoFinnigan Neptune Multi-Collector (MC) Inductively Coupled Plasma Mass Spectrometer (ICPMS). Instrumental biases are assessed and corrected by adopting a standard - sample bracketing procedure to derive correction factors e.g. for mass fractionation effects. U and Th solutions are measured separately; NBL-112a is used for U isotope measurements as the bracketing U-standard and an in-house ²²⁹Th-²³⁰Th-²³²Th standard solution for Th measurements. Further details of our MC-ICPMS procedures can be found in references (9, 10). U-series dating of speleothems is described in more detail in reference (39). Minimum ages are quoted as measured age minus 2 σ and maximum ages as measured age plus 2 σ .

Date Reporting Conventions

Unlike radiocarbon dates, U-series disequilibrium produces results in calendar years. To distinguish between radiocarbon years and U-series results we quote U-series ages as *ky* (thousands of years), uncalibrated radiocarbon dates as ¹⁴C *yr BP* (radiocarbon years before present, the present being the year 1950 AD), and calibrated radiocarbon dates as *cal yr BP* (calibrated years before present, equivalent to calendar years). For dates (i.e. points in time in the past) we use *ka* (thousands of years before today).

Supplementary Figures

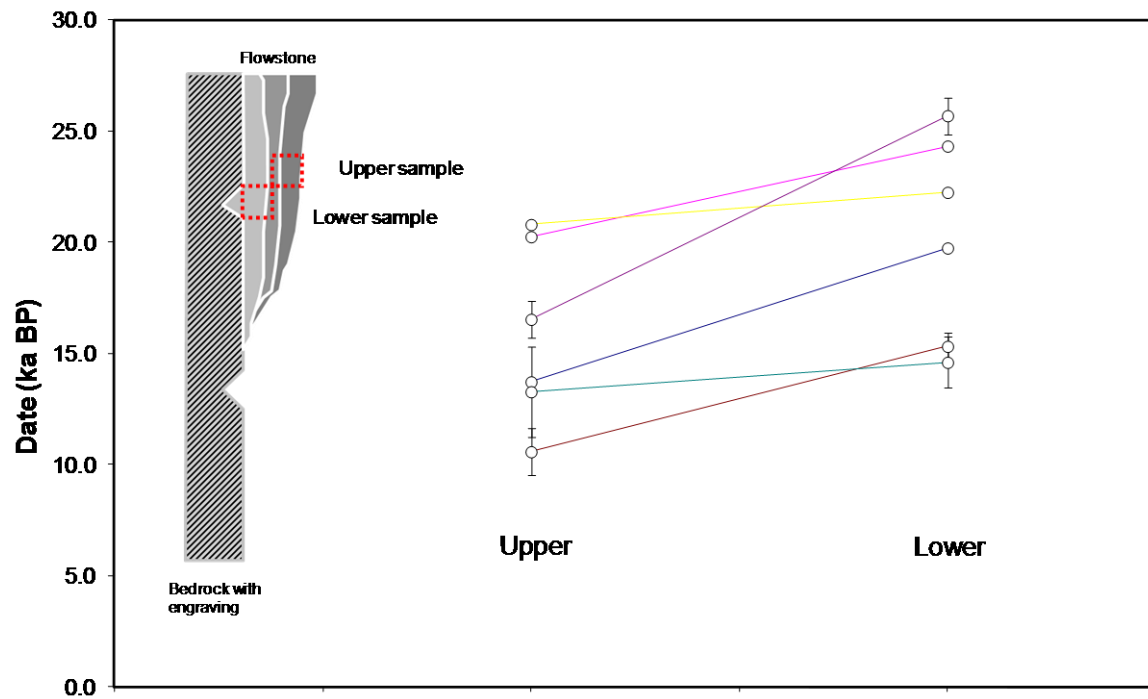


Fig. S1. Uranium series dates on paired samples. Aliquots of samples were removed, representing the upper and lower portions of the calcite crust and dated separately. In all cases, the dates of the upper portions are younger than the lower portions (i.e. following stratigraphic deposition of the calcite), demonstrating the integrity of the samples.



Fig. S2. Sample O-53, overlays red ‘spotted outline’ horse of *Techo de los Polícromos* chamber, Altamira Cave. The location of this symbol on the *Techo de los Polícromos* is shown in Fig. S9. Image ©National Museum and Research Centre of Altamira / Pedro Saura.



Fig. S3. Sample O-80, El Castillo Cave, overlays black outline drawing of an indeterminate animal in corridor of *Techo de las Manos*.

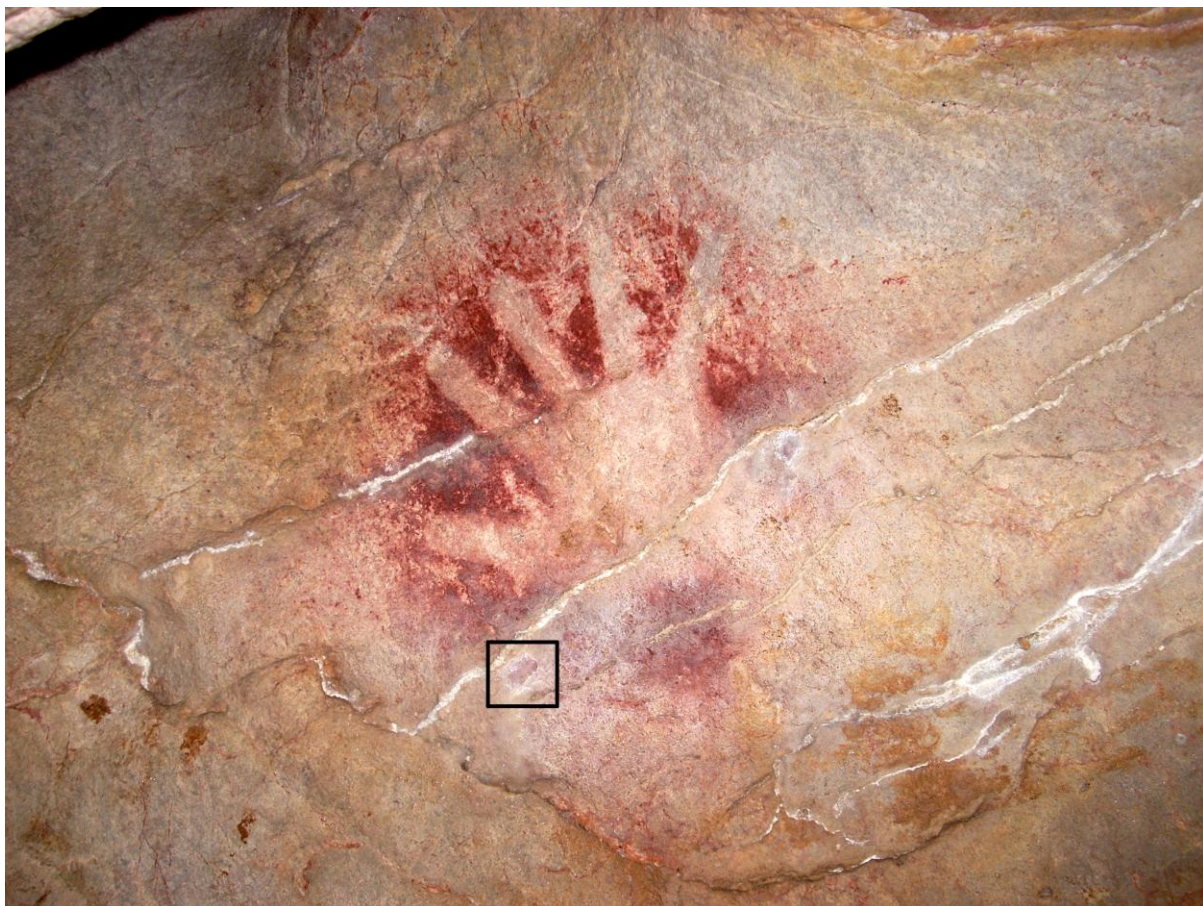


Fig. S4. Sample O-58 overlays red stippled negative hand stencil of *Techo de las Manos*, El Castillo Cave. Note red pigment revealed under sample.



Fig. S5. Anthropomorph figure of *Galería de los Antropomorfo*, Tito Bustillo Cave. Sample O-58 overlays red pigment of vertex of scarf stalactite; sample O-48 is drilled from a recent break in the stalactite providing a maximum age for the figure.

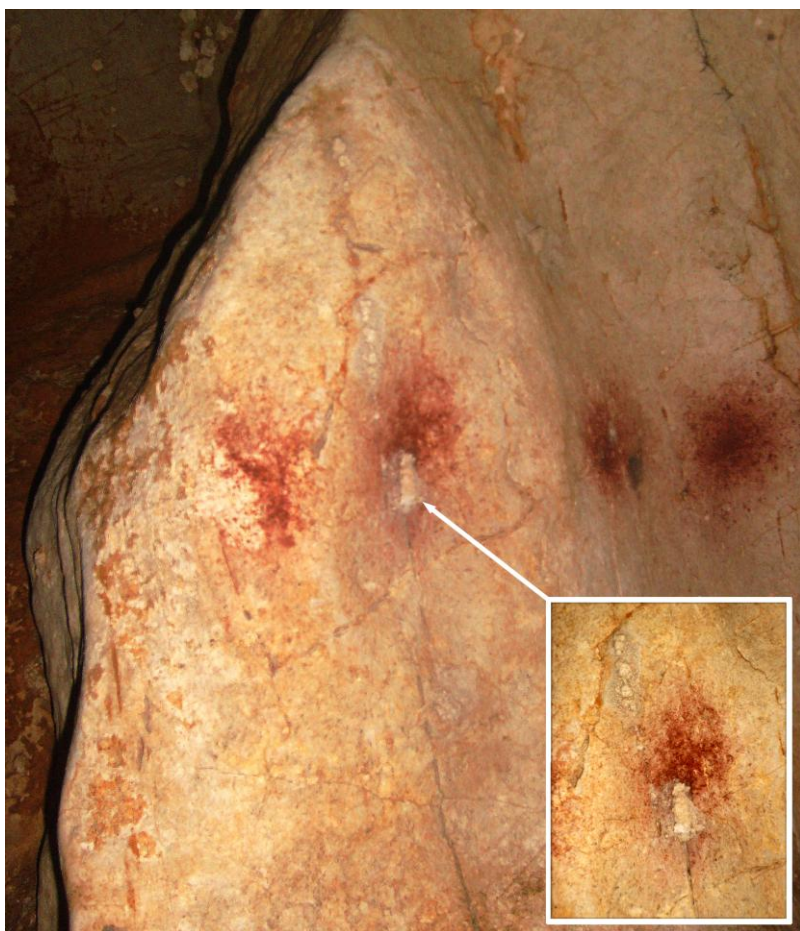


Fig. S6. Sample O-69 overlays large red disk of *Galería de los Discos*, El Castillo Cave.

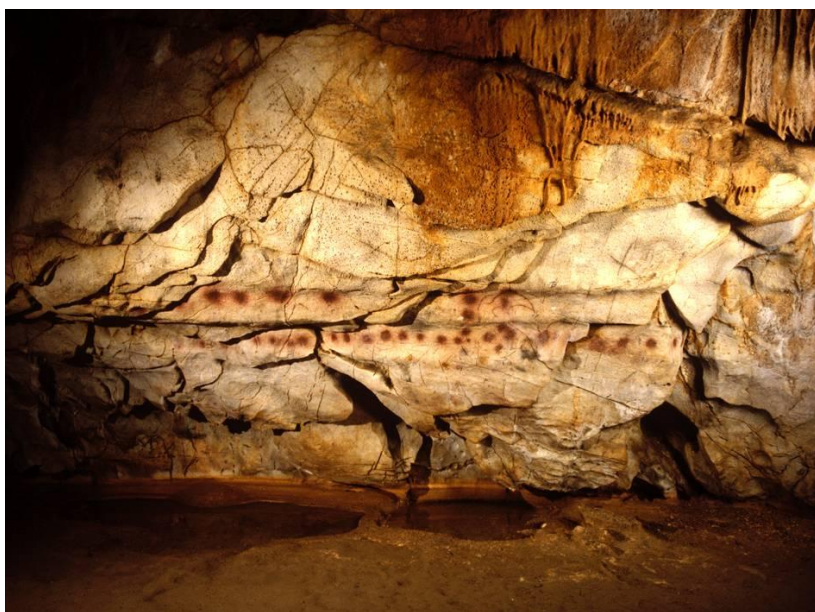


Fig. S7. *Galería de los Discos*, El Castillo Cave. Sample O-87 underlies a large red disk, and provides a maximum age. (Image © Consejería de Cultura, Turismo y Deporte, Gobierno de Cantabria / Pedro Saura)



Fig. S8. Sample O-50 overlays large red claviform-like symbol on the *Techo de los Polícromos*, Altamira Cave. The location of this symbol on the *Techo de los Polícromos* is shown in Fig. S9. Image © The National Museum and Research Centre of Altamira / Pedro Saura.

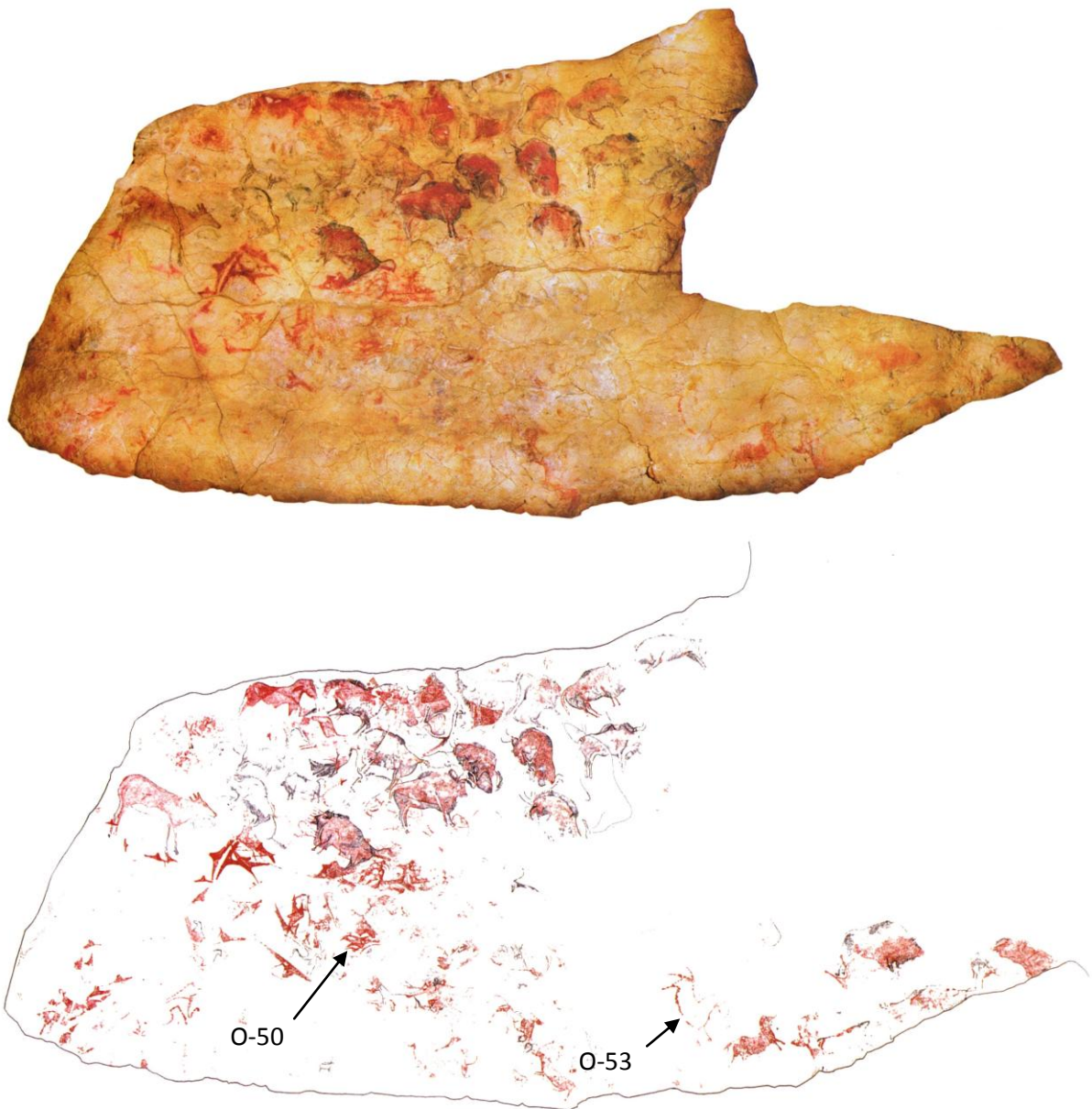


Fig. S9. Digital reconstruction (top) and drawing (bottom) of the *Techo de los Polícromos*, Altamira Cave, showing the location of the claviform-like symbol (sample O-50) and the red spotted outline horse (sample O-53). Images ©The National Museum and Research Centre of Altamira / Pedro Saura.

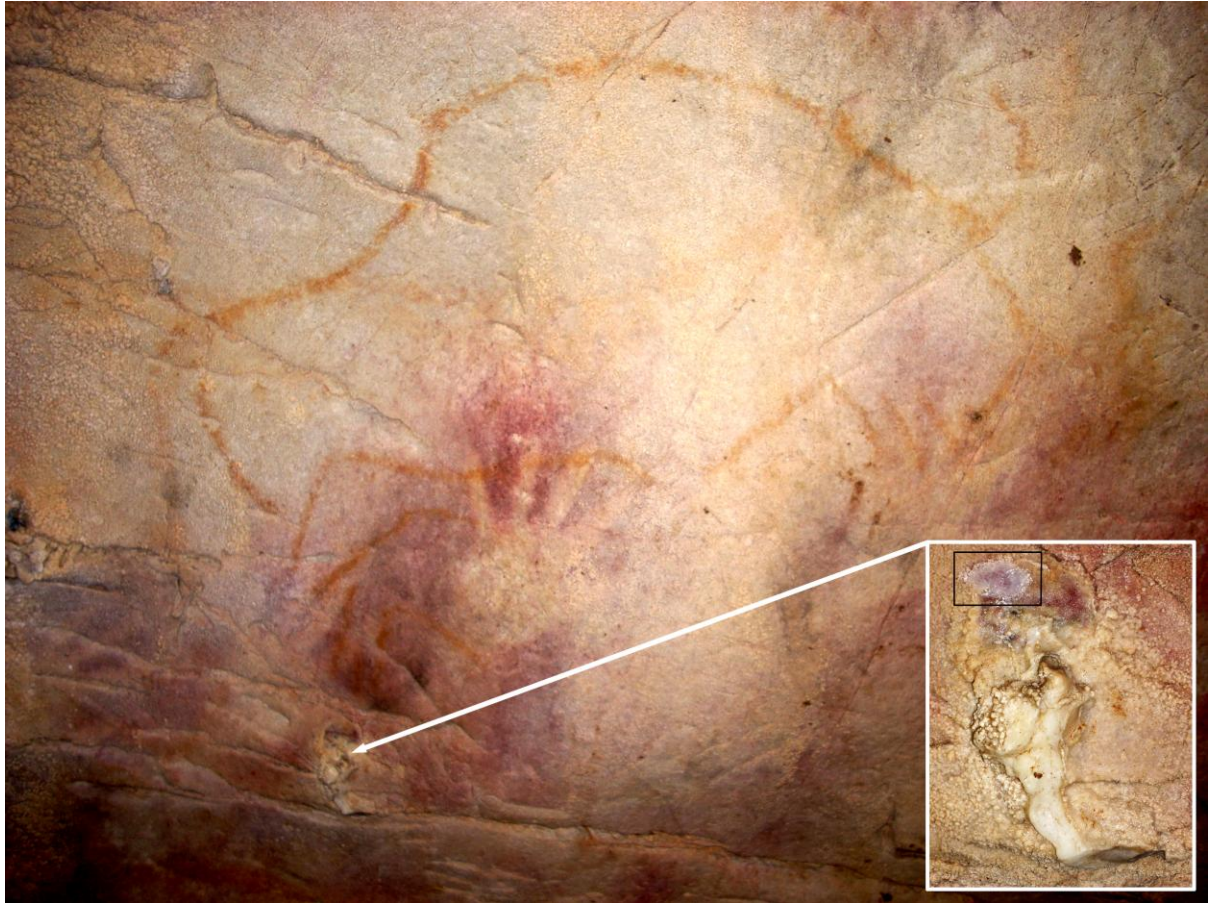


Fig. S10. Sample O-82 overlays red negative hand stencil, and underlies yellow outline bison of *Panel de las Manos*, El Castillo Cave. See also Fig. S12. Note the red pigment revealed under the sample.



Fig. S11. Sample O-83 overlays large red stippled disk on the *Panel de las Manos*, El Castillo Cave. The age of >40.8 ky makes this the oldest known cave art in Europe. The pre-Gravettian date for a hand stencil on the same panel (O-82 at >37.3 ky) and the similarity in painting technique may indicate that all the stippled paintings on this panel are contemporary representing more than 50 motifs (see Fig. S12). The yellow bison is superimposed on this composition and represents a later addition to the panel.

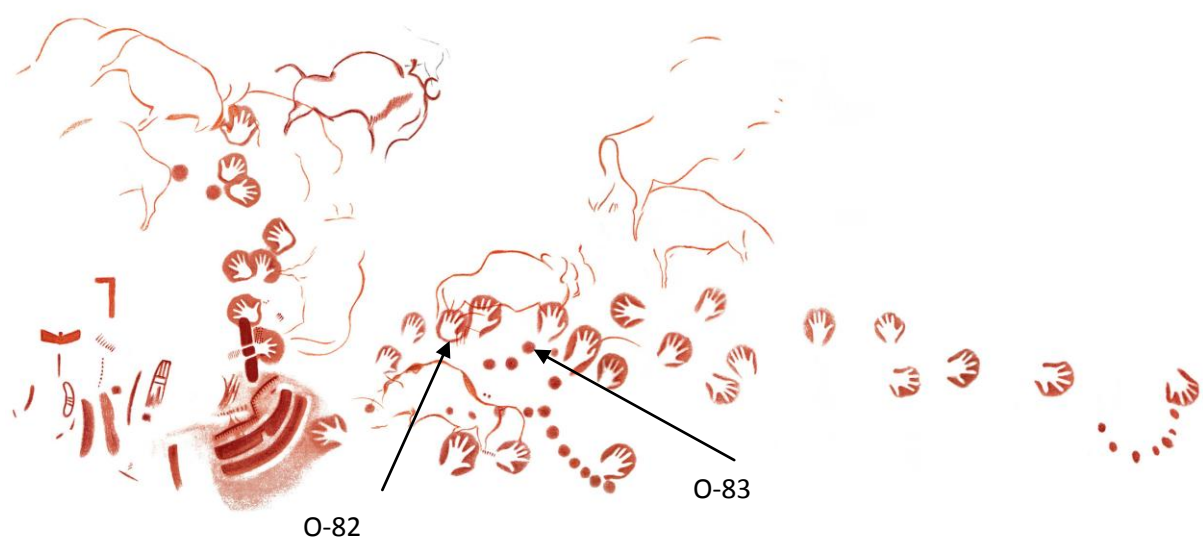


Fig. S12. The *Panel de las Manos*, El Castillo Cave showing the location of samples O-82 overlaying a negative hand stencil, >37.3 ky , and O-83 overlaying a large red stippled disk, >40.8 ky . The tracing in the lower panel is taken from ref (35).

Sample BIG-UTh-	Site	Description	$^{230}\text{Th}/^{238}\text{U}$	$^{234}\text{U}/^{238}\text{U}$	$^{230}\text{Th}/^{232}\text{Th}$	Uncorrected Age (ky)	Corrected Age (ky)
O-30	Tito Bustillo	Overlies red horse, <i>Ensemble X</i>	0.001521 ± 0.000042	0.8791 ± 0.0016	9.60 ± 0.29	0.1891 ± 0.0053	0.1734 ± 0.0095
O-101	La Pasiega	Overlies red bovid, Pasiega C	0.01962 ± 0.00047	2.1559 ± 0.0043	2.959 ± 0.075	0.998 ± 0.024	0.73 ± 0.14
O-103	La Pasiega	Overlies red megaloceros, Pasiega B	0.04866 ± 0.00043	3.0720 ± 0.0058	40.17 ± 0.47	1.741 ± 0.016	1.706 ± 0.023
O-109	La Pasiega	Overlies red undetermined figure, Pasiega B	0.13146 ± 0.00066	6.384 ± 0.010	225.5 ± 1.9	2.266 ± 0.012	2.258 ± 0.013
O-88	El Castillo	Overlies small red dot, <i>Gran Sala</i>	0.08653 ± 0.00049	4.0458 ± 0.0072	116.03 ± 0.81	2.355 ± 0.014	2.339 ± 0.016
O-106	La Pasiega	Overlies red undetermined figure, Pasiega B	0.14250 ± 0.00090	6.213 ± 0.011	695.5 ± 5.9	2.526 ± 0.017	2.523 ± 0.017
O-71	Altamira	Overlies black ibex, <i>La Hoya</i>	0.05320 ± 0.00081	1.6567 ± 0.0030	3.964 ± 0.058	3.557 ± 0.055	2.85 ± 0.35
O-107	La Pasiega	Overlies red bison, Pasiega B	0.1557 ± 0.0011	5.0341 ± 0.0080	24.42 ± 0.27	3.417 ± 0.025	3.307 ± 0.055
O-108	La Pasiega	Overlies red bison, Pasiega B	0.1308 ± 0.0010	4.2912 ± 0.0075	100.58 ± 0.96	3.368 ± 0.027	3.342 ± 0.029
O-105	La Pasiega	Overlies red horses, Pasiega B	0.11682 ± 0.00065	3.2383 ± 0.0055	106.09 ± 0.96	3.996 ± 0.024	3.967 ± 0.027

O-110	La Pasiega	Overlies red horse, Pasiega B	0.3140 ± 0.0019	7.857 ± 0.014	38.91 ± 0.34	4.429 ± 0.029	4.340 ± 0.044
O-73	La Pasiega	Overlies red triangular symbol, Pasiega C	0.13997 ± 0.00062	3.3805 ± 0.0054	308.1 ± 1.8	4.596 ± 0.022	4.585 ± 0.022
O-102	La Pasiega	Overlies black ibex, Pasiega C	0.1058 ± 0.0011	2.1715 ± 0.0040	38.59 ± 0.45	5.433 ± 0.059	5.323 ± 0.078
O-76	La Pasiega	Overlies red claviform, Pasiega B	0.2330 ± 0.0045	4.5823 ± 0.019	92.6 ± 1.9	5.66 ± 0.11	5.615 ± 0.116
O-46	Altamira	Overlies red techtiform, sector III	0.07980 ± 0.00047	1.4959 ± 0.0026	40.29 ± 0.35	5.969 ± 0.038	5.854 ± 0.068
O-84	El Castillo	Overlies red deer, <i>Galería del Bisonte</i>	0.1068 ± 0.0022	1.4707 ± 0.0027	2.812 ± 0.061	8.20 ± 0.18	5.9 ± 1.1
O-77	Covalanas	Overlies red bovid	0.1046 ± 0.0011	1.8566 ± 0.0037	79.49 ± 0.86	6.304 ± 0.067	6.242 ± 0.073
O-78	Santián	Overlies red “hand-like” symbol	0.07676 ± 0.00051	1.3083 ± 0.0027	140.4 ± 1.1	6.588 ± 0.047	6.552 ± 0.050
O-22	Tito Bustillo	Red pigment associated with anthropomorphic figure, <i>Galería de los Antropomorfos</i>	0.1872 ± 0.0027	1.9757 ± 0.0040	2.115 ± 0.031	10.79 ± 0.17	6.9 ± 1.9
O-98	La Pasiega	Overlies small red dot, Pasiega C	0.2337 ± 0.0012	3.6645 ± 0.0065	155.8 ± 1.3	7.142 ± 0.041	7.107 ± 0.044
O-68	El Castillo	Overlies black horse, <i>El Paso</i>	0.3543 ± 0.0026	4.9939 ± 0.0090	15.92 ± 0.10	7.964 ± 0.063	7.58 ± 0.16
O-56	Covalanas	Overlies red deer	0.13612 ± 0.00099	1.8976 ± 0.0034	26.26 ± 0.24	8.083 ± 0.062	7.85 ± 0.13

O-60	Santián	Overlies red color concentration on stalagmitic pillar, Main Corridor	0.1490 ± 0.0029	1.9056 ± 0.0077	9.10 ± 0.20	8.84 ± 0.18	8.09 ± 0.39
O-91	El Castillo	Overlies black bovid, <i>Galería del Bisonte</i>	0.3136 ± 0.0028	4.1893 ± 0.0088	57.66 ± 0.58	8.418 ± 0.080	8.306 ± 0.091
O-74	La Pasiega	Overlies yellow double arch motif, Pasiega C	0.3010 ± 0.0013	3.8029 ± 0.0064	36.62 ± 0.12	8.918 ± 0.041	8.732 ± 0.085
O-100	La Pasiega	Overlies red deer, Pasiega C	0.2881 ± 0.0019	3.6286 ± 0.0078	281.5 ± 2.3	8.948 ± 0.063	8.924 ± 0.063
O-89	El Castillo	Overlies red 'bell', <i>Panel de los Campaniformes</i>	0.15478 ± 0.00097	1.8411 ± 0.0033	62.44 ± 0.49	9.528 ± 0.064	9.412 ± 0.084
O-85	El Castillo	Overlies red rectangular motif, <i>Galería del Biosnte</i>	0.2579 ± 0.0064	2.2827 ± 0.0065	3.446 ± 0.092	12.95 ± 0.34	10.1 ± 1.3
O-23	Tito Bustillo	Overlies red vulva, <i>Cámara de las vulvas</i>	0.2153 ± 0.0016	1.6909 ± 0.0035	3.021 ± 0.021	14.74 ± 0.12	11.1 ± 1.7
O-97	La Pasiega	Overlies red deer, Pasiega C	0.2940 ± 0.0022	2.6004 ± 0.0053	9.177 ± 0.067	12.95 ± 0.10	11.89 ± 0.45
O-17	Tito Bustillo	Overlies violet horse, <i>Ensemble IX</i>	0.11036 ± 0.00061	0.8731 ± 0.0014	4.828 ± 0.022	14.803 ± 0.092	12.5 ± 1.2
O-99	La Pasiega	Overlies red dot, Pasiega C	0.3894 ± 0.0024	3.4601 ± 0.0060	33.86 ± 0.16	12.863 ± 0.085	12.58 ± 0.14
O-40	Las Aguas	Overlies red and	0.13338 ± 0.00070	1.1321 ± 0.0019	17.652 ± 0.075	13.656 ± 0.080	13.07 ± 0.30

		engraved bison, Principal Panel					
O-14	Tito Bustillo	Overlies red horse, <i>Ensemble X</i>	0.2424 ± 0.0015	1.6578 ± 0.0032	5.128 ± 0.026	17.09 ± 0.12	14.6 ± 1.1
O-86	El Castillo	Overlies black bison, <i>El Paso</i>	0.5580 ± 0.0078	3.6635 ± 0.0087	4.931 ± 0.084	17.70 ± 0.27	15.06 ± 0.99
O-12	Tito Bustillo	Red horse head, <i>Ensemble X</i>	0.2346 ± 0.0017	1.6474 ± 0.0035	9.595 ± 0.063	16.61 ± 0.14	15.33 ± 0.60
O-9	Tito Bustillo	Red horse, <i>Ensemble X</i>	0.1112 ± 0.0010	0.7366 ± 0.0018	9.027 ± 0.088	18.05 ± 0.19	16.55 ± 0.81
O-67	El Castillo	New growth of broken scarf stalagtite with red disk, <i>Galería del Bisonte</i>	0.2174 ± 0.0015	1.4205 ± 0.0033	14.91 ± 0.13	18.00 ± 0.14	17.11 ± 0.44
O-81	El Castillo	Overlies red disk, <i>Corredor Techo de las Manos</i>	0.6046 ± 0.0044	3.7396 ± 0.0071	27.22 ± 0.25	18.86 ± 0.15	18.36 ± 0.23
O-72	La Pasiega	Overlies red triangle, Pasiega C	0.7673 ± 0.0033	4.8203 ± 0.0090	260.40 ± 0.64	18.519 ± 0.092	18.468 ± 0.094
O-43	Las Aguas	Overlies red quadrangular symbol, Chamber of Engravings	0.2257 ± 0.0010	1.3494 ± 0.0026	181.0 ± 1.1	19.83 ± 0.10	19.75 ± 0.11
O-53	Altamira	Overlies red 'spotted outline' horse, <i>Techo de los Polícromos</i>	0.2884 ± 0.0013	1.5471 ± 0.0026	107.07 ± 0.20	22.26 ± 0.11	22.11 ± 0.13

O-70	Las Aguas	Overlies brown ‘T’ sign, Principal Panel	0.2266 ± 0.0013	1.1772 ± 0.0021	18.03 ± 0.085	23.22 ± 0.16	22.29 ± 0.47
O-80	El Castillo	Overlies black indeterminate animal, <i>Corredor Techo de las Manos</i>	0.7879 ± 0.0047	3.9828 ± 0.0073	30.01 ± 0.15	23.43 ± 0.16	22.88 ± 0.27
O-58	El Castillo	Overlies red negative hand stencil, <i>Techo de las Manos</i>	0.5272 ± 0.0020	2.5774 ± 0.0049	222.70 ± 0.49	24.42 ± 0.11	24.34 ± 0.12
O-21	Tito Bustillo	Red pigment associated with anthropomorphic figure, <i>Galería de los Antropomorfos</i>	0.6252 ± 0.0031	1.8038 ± 0.0037	2.17 ± 0.01	44.94 ± 0.29	29.65 ± 0.55^a
O-69	El Castillo	Red disk, <i>Galería de los Discos</i>	0.7512 ± 0.0029	2.7072 ± 0.0051	788.24 ± 5.5	34.28 ± 0.17	34.25 ± 0.17
O-50	Altamira	Overlies red claviform-like symbol, <i>Techo de los Polícromos</i>	0.4933 ± 0.0024	1.6594 ± 0.0030	17.473 ± 0.068	37.60 ± 0.23	36.16 ± 0.61
O-82	El Castillo	Overlies red negative hand stencil and underlies yellow outlined bison, <i>Panel de las Manos</i>	0.51115 ± 0.0029	1.6970 ± 0.0035	48.81 ± 0.49	38.15 ± 0.27	37.63 ± 0.34
O-83	El Castillo	Overlies large red disk, <i>Panel de las Manos</i>	0.35732 ± 0.0022	1.1048 ± 0.0020	28.64 ± 0.29	42.38 ± 0.33	41.40 ± 0.57

Table S1 Results of the U-series disequilibrium dating of samples of calcite from above the art (and, thus, minimum ages only) plotted in Fig. 2. Isotopic ratios are given as activity ratios, errors are at 2σ . Ages are corrected for detritus using an assumed $^{232}\text{Th}/^{238}\text{Th}$ activity of 1.250 ± 0.625 and $^{230}\text{Th}/^{238}\text{U}$ and $^{234}\text{U}/^{238}\text{U}$ at equilibrium, except (a) which is corrected using measured values on insoluble residue $^{230}\text{Th}/^{232}\text{Th}=0.8561 \pm 0.0039$.

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